

SPATIAL TAXONOMY · PROOF OF CONCEPT

# Building Width Calculator

Furniture → Zone → Width. A backwards-running method for sizing office buildings from tenant fit-out, driving the tool-buildingwidth engine.

## AGENDA — FIVE SLIDES

- 01 The Backwards Method
- 02 Zone Depth by Use Type
- 03 The Width Calculator
- 04 The Floor Plate
- 05 What Comes Next

# The Backwards Method

Most office buildings start with a width — sixty feet for daylight — then fit furniture into what remains. Woodfine inverts the sequence: start with the tenant's desk, workstation, exam table or bench; the zones follow from the furniture, and the building width is whatever the arrangement requires. The cross-section has three zones, with Zone 1 (Habitat) and Zone 2 (Magazine) mirrored on each side of a single centreline Zone 3 (Corridor).

|        |                     |                   |
|--------|---------------------|-------------------|
| Facade | Frontage            | —                 |
| Zone 1 | Habitat — desks     | 6 m               |
| Zone 2 | Magazine — storage  | 3 m               |
| Zone 3 | Corridor            | 3 m               |
| Zone 2 | Magazine — storage  | 3 m               |
| Zone 1 | Habitat — desks     | 6 m               |
| Facade | Frontage            | —                 |
| TOTAL  | Professional Office | 21.0 m<br>68' 11" |

## Zone 1 — Habitat

Desks, workstations, exam tables, benches. Six metres from the façade is the working baseline — the European Lighting Standard threshold for daylight at a workstation; some use types widen it (Medical 7.28 m for exam-table depth) and Academic compresses it because seating faces forward, not toward the façade.

## Zone 2 — Magazine

Storage, bookshelves, filing, staff rooms, server rooms. The slack variable: widen it to balance the floor plate so it doesn't end up as a long, narrow rectangle.

## Zone 3 — Corridor

A single centreline corridor, not mirrored. Width set by egress and accessibility — oversized by 20% to promote well-being and absorb the "in and out" traffic of smaller tenants.

**The 0.7 m German Circulation Law addition.** Perpendicular desks at 6.0 m Habitat fit only two in series; 0.7 m unlocks the third with clearance. Token: `bim.circulation-addition.perpendicular-desk = 0.7 m`.

# Zone Depth by Use Type

Each segment's width is the actual zone depth in metres, scaled against the widest total. Building width = 2×(Habitat + Magazine) + Corridor — the corridor is centreline, single, not mirrored. The widest tenant (Business at its open-storage maximum) is roughly 17.6 m wider than the narrowest (Private Office); a mixed-tenant floor must reconcile this within a single plate width.

Zone 1 — Habitat (desks)

Zone 2 — Magazine (storage)

Zone 3 — Corridor (circulation)

|                                                                                                                |                                                                                                  |                                 |
|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|---------------------------------|
| <b>Private Office</b><br>PO-1/PO-2/PO-3 — opens to shared building corridor (no bilateral corridor)            | <div><div>5.99 m</div><div></div><div></div><div>5.99 m</div></div>                              | <b>14.73 m</b><br>48'4" total   |
| <b>Academic</b><br>Seating faces forward — Habitat under 6 m permitted; opens to building corridor             | <div><div>4.70 m</div><div>3 m</div><div>3 m</div><div>4.70 m</div></div>                        | <b>15.40 m</b><br>50'6" total   |
| <b>Professional Office</b><br>V12 baseline (Jan 2025) — TBD pending furniture lock; Z2 and Z3 are placeholders | <div><div>6 m</div><div>3 m</div><div>3 m</div><div>3 m</div><div>6 m</div></div>                | <b>21.00 m</b><br>68'11" total  |
| <b>Laboratory</b><br>Bench depth + fume hood clearance + IBC high-hazard egress 10'0"                          | <div><div>6.78 m</div><div>4.80 m</div><div>3.05 m</div><div>4.80 m</div><div>6.78 m</div></div> | <b>26.21 m</b><br>86'0" total   |
| <b>Medical</b><br>M3/M1/M2 sketches — 286 5/8" + 192" + 113 7/8"; exam-table depth + patient circulation       | <div><div>7.28 m</div><div>4.88 m</div><div>2.89 m</div><div>4.88 m</div><div>7.28 m</div></div> | <b>27.21 m</b><br>89'3" total   |
| <b>Business</b><br>Notes V3 Option A/A — widest of 21 enumerated options; reflects open storage walls          | <div><div>5.51 m</div><div>9.26 m</div><div>2.75 m</div><div>9.26 m</div><div>5.51 m</div></div> | <b>32.29 m</b><br>105'11" total |
| <b>Civic</b><br>Widest Corridor 12'0" — public assembly egress; awaiting DISCOVERY sketch confirmation         | <div><div>6 m</div><div>7.23 m</div><div>3.60 m</div><div>7.23 m</div><div>6 m</div></div>       | <b>30.06 m</b><br>98'7" total   |

# The Building Width Calculator

Zone 1 + Zone 2 | Zone 3 | Zone 2 + Zone 1

Building Width = 2 × (Habitat + Magazine) + Corridor

| USE TYPE            | Z1 HABITAT      | Z2 MAGAZINE    | Z3 CORRIDOR     | HALF (Z1+Z2) | TOTAL WIDTH       |
|---------------------|-----------------|----------------|-----------------|--------------|-------------------|
| Private Office      | 5.99 m / 19'8"  | 1.37 m / 4'6"  | 0.00 m / –      | 7.37 m       | 14.73 m / 48'4"   |
| Academic            | 4.70 m / 15'5"  | 3 m / 9'10"    | 0.00 m / –      | 7.70 m       | 15.40 m / 50'6"   |
| Professional Office | 6 m / 19'8"     | 3 m / 9'10"    | 3 m / 9'10"     | 9.00 m       | 21.00 m / 68'11"  |
| Laboratory          | 6.78 m / 22'3"  | 4.80 m / 15'9" | 3.05 m / 10'0"  | 11.58 m      | 26.21 m / 86'0"   |
| Medical             | 7.28 m / 23'11" | 4.88 m / 16'0" | 2.89 m / 9'6"   | 12.16 m      | 27.21 m / 89'3"   |
| Business            | 5.51 m / 18'1"  | 9.26 m / 30'5" | 2.75 m / 9'0"   | 14.77 m      | 32.29 m / 105'11" |
| Civic               | 6 m / 19'8"     | 7.23 m / 23'9" | 3.60 m / 11'10" | 13.23 m      | 30.06 m / 98'7"   |

WIDEST BUILDING WIDTH

32.29 m

105'11" · Business

NARROWEST BUILDING WIDTH

14.73 m

48'4" · Private Office

SPREAD — WIDEST MINUS NARROWEST

17.56 m

57.6 ft

**The ±5 ft precision question.** Tiles must compose to a single floor-plate width across all use types on a floor. tool-buildingwidth resolves this by computing each floor's tenant-mix maximum width at design time, then redistributing Zone 2 or constraining the mix.

# The Floor Plate

Tiles are the assembly units — Small Tile = 2,700 SF, Large Tile = 4,900 SF — and they compose to fill the plate exactly. Each Tile is an independent climate zone: a tenant who wants thermostat control leases a Tile. Three example compositions below.

### Composition A — Corporate-heavy

Three large corporate tiles (Tile F, 4,900 SF each) plus two standard corporate tiles (Tile A, 2,700 SF each). Five independent climate zones — each tenant controls their own thermostat. Total: 20,100 SF.

|                                                                 |                                                                 |                                                                 |                                                                 |                                                                 |
|-----------------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------|
| <div>Tile F</div> <div>4,900 SF</div> <div>1 climate zone</div> | <div>Tile F</div> <div>4,900 SF</div> <div>1 climate zone</div> | <div>Tile F</div> <div>4,900 SF</div> <div>1 climate zone</div> | <div>Tile A</div> <div>2,700 SF</div> <div>1 climate zone</div> | <div>Tile A</div> <div>2,700 SF</div> <div>1 climate zone</div> |
|-----------------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------|

Tiles: 5Climate zones: 5Total: 20,100 SF (target 20,000)

### Composition B — Private Office heavy

Maximum climate-zone count. Two Tile G (4,900 SF, 10 climate zones each) plus four Tile B-1 (2,700 SF, 5 climate zones each) end caps. Twenty private licensed offices and twenty more in end caps. Forty independent thermostats on one floor.

|                                                                    |                                                                   |                                                                    |                                                                    |                                                                   |                                                                    |
|--------------------------------------------------------------------|-------------------------------------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------------------|-------------------------------------------------------------------|--------------------------------------------------------------------|
| <div>Tile B-1</div> <div>2,700 SF</div> <div>5 climate zones</div> | <div>Tile G</div> <div>4,900 SF</div> <div>10 climate zones</div> | <div>Tile B-1</div> <div>2,700 SF</div> <div>5 climate zones</div> | <div>Tile B-1</div> <div>2,700 SF</div> <div>5 climate zones</div> | <div>Tile G</div> <div>4,900 SF</div> <div>10 climate zones</div> | <div>Tile B-1</div> <div>2,700 SF</div> <div>5 climate zones</div> |
|--------------------------------------------------------------------|-------------------------------------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------------------|-------------------------------------------------------------------|--------------------------------------------------------------------|

Tiles: 6Climate zones: 40Total: 20,600 SF (target 20,000)

### Composition C — Mixed Professional Centre

Two Tile H (large professional, 3 climate zones each), one Tile C-2 (Medium professional pair), one Tile C-1 (Medium + Small), and two end-cap mixed tiles (E-1 / E-2). Tenants: Medical anchor + Business cluster + Private Office end caps.

|                                                                    |                                                                  |                                                                    |                                                                  |                                                                    |                                                                    |
|--------------------------------------------------------------------|------------------------------------------------------------------|--------------------------------------------------------------------|------------------------------------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------------------|
| <div>Tile E-1</div> <div>2,700 SF</div> <div>3 climate zones</div> | <div>Tile H</div> <div>4,900 SF</div> <div>3 climate zones</div> | <div>Tile C-2</div> <div>2,700 SF</div> <div>2 climate zones</div> | <div>Tile H</div> <div>4,900 SF</div> <div>3 climate zones</div> | <div>Tile C-1</div> <div>2,700 SF</div> <div>2 climate zones</div> | <div>Tile E-2</div> <div>2,100 SF</div> <div>3 climate zones</div> |
|--------------------------------------------------------------------|------------------------------------------------------------------|--------------------------------------------------------------------|------------------------------------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------------------|

Tiles: 6Climate zones: 16Total: 20,000 SF (target 20,000)

Tile = Climate Zone. Composition B (Private Office heavy) yields forty independent thermostats on one floor. Source token: `bim.tile.design-principles.climate-zone-autonomy`.

# What Comes Next

This HTML is the specification document for `tool-buildingwidth`, a Rust engine that reads the DTCG tokens listed at right and computes floor plate dimensions, tile compositions, and demising-wall coordinates from a tenant-mix input.

The engine outputs IFC-compatible geometry and a leasing-plan summary; this deck is what the engine will render as a human preview.

Every metre, every square foot, every percentage on these slides is read live from the DTCG token files at page load. When the tokens change — a new key plan sample, a corridor width update — the deck changes with them. The engine, when shipped, shares the same source.

Source: `woodfine-design-bim` DTCG token library v0.0.2

SOURCE TOKEN FILES

building-width-calculator.dtcg.json  
floor-plate-standards.dtcg.json  
professional-office-subtypes.dtcg.json  
tile-system.dtcg.json

FUTURE CONSUMER

pointsav-monorepo/  
 crates/tool-buildingwidth/